$$= \frac{1.5 \text{ mol}}{37.6 \cancel{\text{dm}^3}} \times 16.5 \cancel{\text{dm}^3}$$

$$= 0.66 \text{ mol}$$

47. n = 1.82 mole

$V = 5.43 \text{ dm}^3$

$T = 69.5 + 273 = 342.5$

P = ?

PV = nRT

$$P = \frac{nRT}{V}$$

$$P = \frac{1.82 \text{ mol} \times 0.0821 \cancel{\text{dm}^3} \text{ atm mol}^{-1} \text{ K}^{-1} \times 342.5 \text{ K}}{5.43 \cancel{\text{dm}^3}}$$

$$P = 9.425 \text{ atm}$$

48.

$P_1 = 1.2$ atm

$T_1 = 18^0$ C + 273 = 291 K

$T_2 = 85^0$ C + 273 = 358 K

P_2 = ?

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$P_2 = \left(\frac{P_1}{T_1}\right) \times T_2$$

$$= \frac{1.2 \text{ atm}}{291 \cancel{K}} \times 358 \cancel{K}$$

$$P_2 = 1.48 \text{ atm}$$

49. $T_1 = 6\ ^0$C + 273 = 279 K

$P_1 = 4$ atm $\quad V_1 = 1.5$ ml

$T_2 = 25^0$ C + 273 = 298 K

$P_2 = 1$ atm $\quad V_2 = ?$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$V_2 = \frac{P_1V_1}{T_1} \times \frac{T_2}{P_2}$$

$$= \frac{4 \cancel{\text{atm}} \times 1.5 \text{ ml} \times 298 \cancel{K}}{279 \cancel{K} \times 1 \cancel{\text{atm}}}$$

$V_2 = 6.41$ ml

50. Given,

$V = 154.4 \times 10^{-3} \text{ dm}^3$,

P = 742 mm of Hg

T = 298 K $\quad$ m = ?

$$n = \frac{PV}{RT} = \frac{742 \text{ mm Hg} \times 154.4 \times 10^{-3} \text{ L}}{62 \text{ mm Hg L K}^{-1}\text{mol}^{-1} \times 298 \text{ K}}$$

$$= 0.006 \text{ mol}$$

$$n = \frac{\text{Mass}}{\text{Molar mass}}$$

$$\text{Mass} = n \times \text{Molar mass}$$

$$= 0.006 \times 2.016$$

$$= 0.0121 \text{ g} = 12.1 \text{ mg}$$

51.

$$\frac{\gamma_{unknown}}{\gamma_{N_2}} = \frac{t_{N_2}}{t_{unknown}} = \sqrt{\frac{M_{N_2}}{M_{unknown}}}$$

$$\frac{84 \text{ sec}}{192 \text{ sec}} = \sqrt{\frac{28 \text{ g mol}^{-1}}{M_{unknown}}}$$

On squaring both sides and rearranging

$$\left(\frac{84 \text{ sec}}{192 \text{ sec}}\right)^2 = \frac{28 \text{ g mol}^{-1}}{M_{unknown}}$$

$$M_{unknown} = 28 \text{ g mol}^{-1} \times \left(\frac{192 \text{ sec}}{84 \text{ sec}}\right)^2$$

$$M_{unknown} = 146.28 \text{ g mol}^{-1}$$

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